Air-conditioning

Domestic air-conditioning (AC or A/C) units are designed to draw in warm air from a living space and then cool it by evaporation, in a similar process to that used in refrigerators. A closed circuit of refrigerant, moved around by a pump, cools warm air drawn in by a fan. The refrigerant then transports the transferred heat out of the building to an external condenser, where the heat dissipates into the air outside. The refrigerant starts another cycle by flowing through an expansion valve, which lowers its pressure and temperature, ready to cool more air. As the air indoors cools, droplets of water vapor condense into liquid, leaving the air less humid as well as cooler.

Home AC unit
An AC unit consists of indoor and outdoor components. The indoor part draws in warm air and cools it; the outdoor part expels the air's heat.

FANS AND AIR CONDITIONERS ACCOUNT FOR ABOUT 15 PERCENT OF US ELECTRICITY CONSUMPTION



REFRIGERANTS

These substances change easily between gas and liquid as their temperature changes. As a liquid changes to a gas, the liquid that is left has less energy and becomes colder. Chlorofluorocarbons (CFCs) were widely used as refrigerants until they were found to damage the atmosphere's ozone layer. Hydrofluorocarbons (HFCs) are now used in household appliances.



